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Review

Gut microbiota variation and diversity and gut graft-versus-host disease (GVHD) in pediatrics: A systematic review

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A R T I C L E I N F O

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A B S T R A C T

Background: Allogeneic hematopoietic stem cell transplantation (HSCT) provides children with life-threatening conditions an opportunity for survival. Complications from graft-versus-host disease (GVHD) are a major source of morbidity and death, recently linked to gut dysbiosis in the hematopoietic stem cell transplantation (HSCT) population. But so far, no comprehensive study has been conducted to investigate this relationship in the children population. In this systematic study, we investigated the Gut microbiota variation and diversity and gut GVHD in pediatrics.
Methods: A systematic review according to PRISMA standards was performed from inception till August 2024. Out of 568 originally chosen publications, 10 studies involving 490 pediatric subjects satisfied the eligibility criteria and were included.
Results: The findings obtained from the study included in the present systematic study mostly indicated the use of combined treatments including Busulfan, Cyclophosphamide, and total body irradiation and in some studies the use of anti-thymocyte globulin and Melphalan as conditioning regimens. In addition, out of 10 reviewed studies, 9 reported a significant decrease in gut microbiota diversity following GVHD. However, in all studies, an increased variation was reported. So that most of the studies showed a decrease in the levels of beneficial bacteria and producers of short-chain fatty acid products in the intestine such as *Ruminococcaceae* and *Enterococcus*, which is also observed in the intestinal microbiota population of healthy people.
Conclusion: As a result, our findings indicated a decrease in diversity as well as a change in intestinal microbiota in children with GVHD under HSCT in most of the studies.

1. Introduction

Allogeneic hematopoietic stem cell transplantation (HSCT) offers children with both malignant and non-malignant disorders a chance for survival. In the first year after HSCT, problems from infections and graft-versus-host disease (GVHD) mostly account for death, with one often triggering the other [1]. Acute GVHD occurs during the first 100 days after transplantation, caused by alloreactive donor T cells targeting the recipient's cells, leading to cytokine release and parenchymal injury in organs such as the skin, liver, and gastrointestinal tract [2]. The primary treatment for gut GVHD is systemic glucocorticoids to mitigate the inflammatory response; nevertheless, the failure rate is 50 % [3]. Certain researchers have examined the gut microbiota to ascertain its role in the immunological modulatory pathways causing GVHD.

Numerous studies have shown a correlation between microbial makeup and function and illness and death [4]. Research conducted in the 1970s indicated that the cleansing of intestinal microorganisms reduced the incidence of gut GVHD [4]. Recently, a more complex and

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comprehensive understanding of microbial populations and gut GVHD has developed. The reduction of obligatory anaerobes, such as *Clostridium*, and the mono-dominance of other bacteria, such as *Enterococcus*, are correlated with severe gut graft-versus-host disease (stages 3–4) [5–7]. Adult patients exhibiting diminished intestinal bacterial diversity or mono-dominance of species like *Enterococcus* had elevated overall mortality rates post-HSCT [8,9].

The studies conducted in the field of microbiome and GVHD are very limited, most of the studies show differences in the amount and type of bacterial strains along with changes in intestinal bacterial dysbiosis [10,11]. For example, in one study, a group of bacteria producing propionic acid was less frequent in children with GVHD than in children without GVHD [11]. These bacteria can contribute to the mucosal healing rate of patients with GVHD by changing intestinal integrity. Therefore, it seems that there are deep and challenging relationships between microbial profiles in terms of variation and diversity in the intestines of children with HSCT, which can help identify high-risk children and even new therapeutic interventions.

In view of this issue and also in view of the fact that according to our knowledge, there is no comprehensive study to investigate the relationship between intestinal microbiota and intestinal GVHD in this age group of patients, in this systematic study, we aim to better understand and expand information in this field.

2. Material and methods

2.1. Search strategy

This systematic review was performed without regard to time or language limitations following the Preferred Reporting Criteria for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [12]. The general search principles of the articles included in this study also considering the medical subject headings (MeSH) and Emtree (Embase subject title) in accordance with the following search guidelines for a comprehensive search in all major databases including PubMed/MEDLINE, Web of Science, SCOPUS, and Embase was completed by June 2024: ("Gastrointestinal Microbiome"[Mesh] OR "Microbiota" [Mesh] OR "Microbiome*"[All Fields] OR "Gut Microbiome" [All Fields] OR "Gut Microflora" [All Fields] OR "Gut Microbiota" [All Fields] OR "Gastrointestinal Flora" [All Fields] OR " Gut Flora" [All Fields] OR " Gastrointestinal Microbiota" [All Fields] OR " Intestinal Microbiome" [All Fields] OR "Intestinal Microbiota" [All Fields] OR "Intestinal Microflora" [All Fields] OR "Intestinal Flora" [All Fields) AND ("Graft vs Host Disease"[Mesh] OR " Graft vs Host Disease "[All Fields] OR " GVHD "[All Fields] OR "allogeneic stem cell transplantation"[All Fields] OR allo-HSCT [All Fields]) AND ("Child"[Mesh] OR child*[All Fields] OR "Adolescent"[Mesh] OR Adolescen*[All Fields] OR "Pediatrics"[Mesh] OR Pediatric*[All Fields] OR youth*[All Fields] OR teen*[All Fields]). In addition, to avoid ignoring or losing articles related to the present study, we manually reviewed and searched the lists of all general articles obtained from databases as well as meta-analysis and review articles related to the purpose of the study.

2.2. Eligibility criteria

The papers that fulfilled the specified criteria were chosen for comprehensive text evaluation: 1) Observational research (Cohort, Cross-sectional, or Case-Control) that have investigated the correlation and influence of gut microbiota variance and diversity on gut graft-versus-host disease (GVHD); 2) Research involving juvenile participants undergoing Hematopoietic Cell Transplantation (HCT) ($\leq$18 years old); 3) Studies that provided enough data on the assessed outcomes. Articles were eliminated if they: 1) Were other types of observational studies or clinical trials; 2) Involved adults, pregnant, or lactating women; 3) Did not provide enough data on the assessed outcomes in the intervention groups; 4) Were case reports.

2.3. Data extraction

Two investigators independently reviewed the eligible papers and the following data were extracted from the papers: the first author's name, the study location, the publication year, indication for HCT, the participants' characteristics (gender, age and cancer type), the sample size, conditioning regimen/ stem cell source, dose and type of antibiotics Intervention, Graft source, HCT duration, the GVHD stage, the results and the limitations of the analyzed studies.

2.4. Quality assessment

Table 2 provides the specifics of the study quality assessment. The quality of the included publications was independently evaluated by two writers using the Newcastle-Ottawa Quality Assessment Scale [13]. In this context, studies were categorized as poor (< 5 stars), moderate (5–7 stars), and high quality (> 7 stars).

3. Results

This systematic review included 10 studies [11,7,14–21] and excluded the remaining studies (Fig. 1). The total number of patients in this systematic review was 490 pediatrics. Tables 1 and 2 reveals the characteristics and results of the pooled articles. In addition, the quality results of the articles are shown in Table 3. According to our research, 6 studies have been conducted and published in Europe, 2 articles in America, and 2 articles in Asia between 1990 and 2023. Most patients who developed GVHD had acute lymphoblastic and acute myeloid leukemia. The conditioning regimens in these studies received by patients were mostly combination therapies. The subjects in the research conducted by Ashley et al. underwent a combinatorial conditioning regimen of Cyclophosphamide (CY), Cytarabine, total body irradiation (TBI), Busulfan (BU), Fludarabine, and anti-thymocyte globulin (ATG). In a research conducted by Biagi et al. with 36 pediatric patients, 88.9 % had myeloablative conditioning, whereas 11.1 % got a non-myeloablative conditioning regimen. Thirty-two patients (91.4 %) underwent a myeloablative conditioning regimen based on BU or treosulfan, three patients (8.6 %) got a conditioning regimen based on TBI, and one patient was administered a non-myeloablative preparative regimen using CY/ fludarabine. Another study by Biagi et al. involved 10 pediatric patients who received thiotepa, CY, fludarabine, or melphalan (MEL). In the study by Luo et al., all patients received myeloablative intensity therapy, such as BU, CY, ATG, as conditioning regimens. In the study by Margolis et al., based on intensity conditioning, 49 (66.2 %), 1 (1.4 %), and 24 (32.4 %) patients received reduced-intensity, non-myeloablative, and myeloablative therapies, respectively. Furthermore, based on the conditioning radiation, 31 (41.9 %) patients did not receive any radiation, 26 (35.1 %) patients received total lymphoid irradiation, and 17 (23 %) patients received TBI. In the study by Ingham et al., 6 (20.68 %) patients received TBI + CY/TBI + VP16, 6 (20.68 %) patients received combinations of BU, CY, VP16, and MEL, 12 (41.38 %) patients received fludarabine-thiotepa combinations, and 5 (17.24 %) patients received fludarabine-thiotepa combination therapy. In contrast, all participants in the study conducted by Vossen et al. (2014) underwent pretreatment with a myeloablative conditioning regimen that included combination chemotherapy agents such as cyclophosphamide, cytosine arabinoside, or etoposide in certain instances, along with total body irradiation (TBI) for children over 2 years old; younger children received busulfan (20 mg/kg orally, total dose) or busulfex (16 mg/kg IV, total dose) in lieu of TBI, and cyclophosphamide (200 mg/kg IV, total dose). They obtained a T cell-replete bone marrow donation from a matched sibling donor (MSD). In another study by Vossen et al. (1990), children with severe aplastic anemia and Fanconi's syndrome received either CY alone, CY plus procarbazine and rabbit ATG, or CY plus TBI. Children with myelodysplastic syndrome and leukemia received a conditioning regimen consisting of plus TBI, with doses depending on their ages. 59 patients

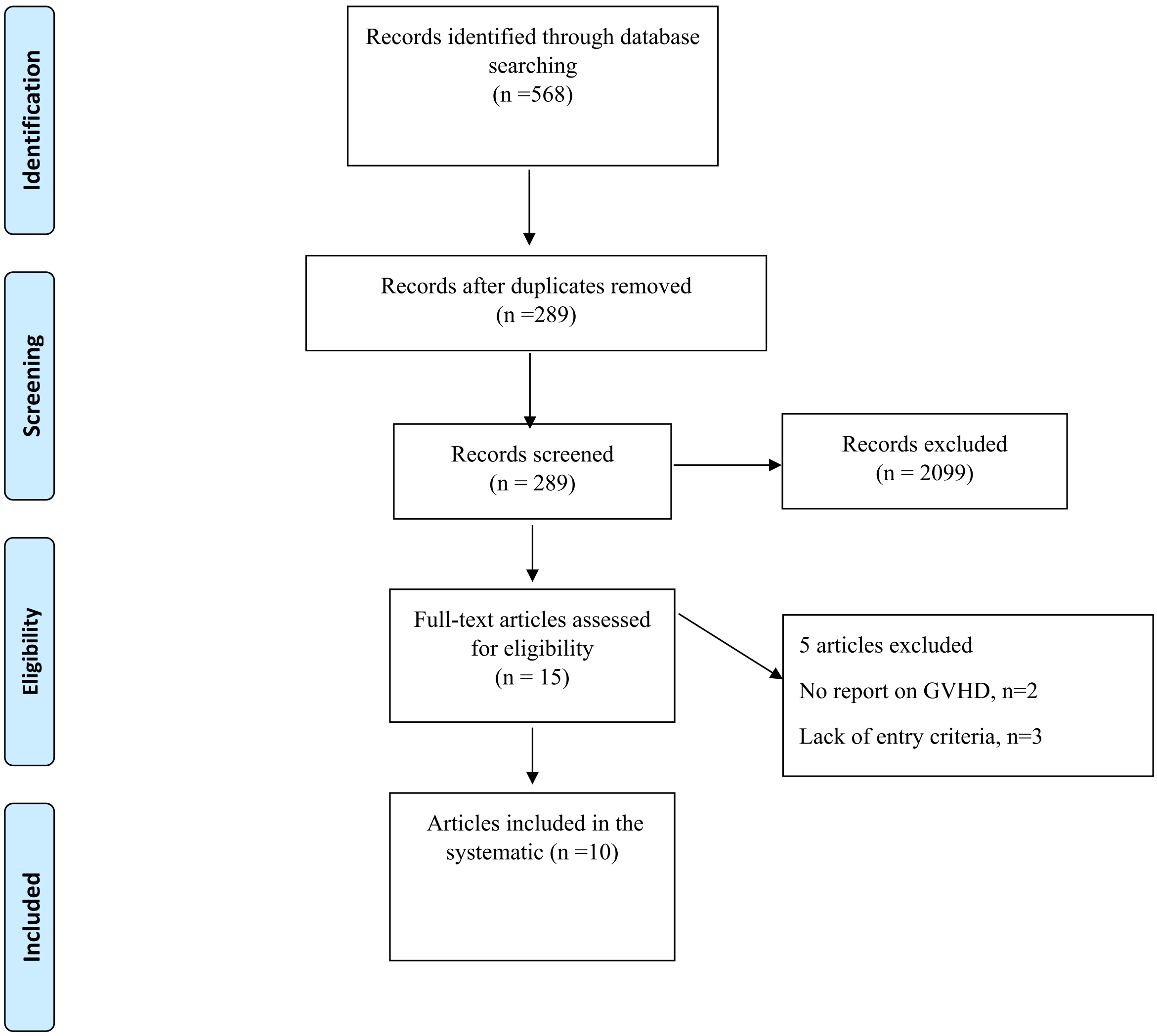

Fig. 1. Flow-chart depicting the study selection process.

received methotrexate for GVHD-prophylaxis for over 100 days after BMT and 6 patients received a continuous intravenous infusion of cyclosporin-A for 2 months followed by oral doses until at least 6 months after BMT.

Concerning the stem cell source, participants in only three studies received stem cell transplants from the bone marrow (studies 3, 9, and 10 in Table 1). The source of stem cell transplant in participants in the Luo et al. study was peripheral blood, whereas in the study by Margolis et al., 27 (36.5 %) patients received stem cell transplants from bone marrow and 47 (63.5 %) patients received it from apheresis. Gray et al. received stem cell transplants from different sources, including matched related donors (MRD), MSD, peripheral blood stem cells (PBSC), bone marrow (BM), umbilical cord blood (UCB), and TBI. Similarly, in the study by Biagi et al., 29 (80.55 %) participants received stem cell transplants from unmanipulated bone marrow cells, 6 (16.66 %) patients received T-cell-depleted peripheral blood stem cells and one patient (2.77 %) received it from cord blood cells. In a study by Ingham et al., the source of stem cell transplant was bone marrow in 23 patients and umbilical cord blood in 6 patients.

A research by Gray et al. shown that stool microbiota variance over time was considerably greater in patients with stage 3–4 gut GVHD compared to the control group (p = 0.007), but pre-HCT microbiota diversity showed no significant difference, indicating gut microbiota stability (p = 0.05) [14]. Decreased variety, reduced *Blautia* levels, and heightened *Fusobacterium* abundance prior to HSCT led to the onset of gut aGVHD in the subjects of the research conducted by Biagi et al. (2019) [19]. In a further research by the same author, non-aGVHD patients exhibited a greater prevalence of propionate-producing *Bacteroidetes* in the pre-HSCT samples [11]. The development of aGVHD correlated with a reduction in the amounts of beneficial *Faecalibacterium* and *Enterococcus*. The study by Vossen et al. (2014) indicated that effective total gastrointestinal decontamination of the graft recipient inhibits moderate to severe acute graft-versus-host disease (aGVHD), likely due to the significant translocation of gastrointestinal microorganisms acting as microbial-associated molecular patterns (MAMPs), which activate macrophages and dendritic cells through pattern-recognition receptors (PRRs) inhibition [20]. This triggers the inflammatory process, resulting in the suppression of acute GVHD. Another significant result of these studies was that gut microbiota diversity was diminished during the occurrence of aGVHD, which correlated with increased mortality and resistance to therapy [20,21]. *Pasteurellaceae* significantly influenced the severity of intestinal aGVHD and diarrhea. *Bifidobacteriaceae* resulted in infectious diarrhea during HSCT. Particular bacteria served as indicators for survival, including *Moraxellaceae*, *Enterococcaceae*, and *Alphaproteobacteria* from the intestinal microbiota post-HSCT, as well as *Proteobacteria*, *Gammaproteobacteria*, and *Odoribacteraceae* prior to HSCT in the research conducted by Song et al. [18]. *Bacteroidetes* and *Firmicutes* at the phylum level, together with *Lachnospiraceae* at the family level, were the predominant bacteria identified in the research conducted by Luo et al. before and after HSCT [16]. *Rothia*

Table 1
Summary of characteristics studies.

First author/ Year/Country	Sample Size	Sex	Stage of GVHD	Indication for HSCT	Conditioning Regimen	Dose and type of antibiotics Intervention	HSCT Duration (Time to onset of GVHD)	Stem cell source	Methods of microbiome analysis
1-N. Gray/ 2023/USA	9 pediatric patients	NA	Maximun Gut GVHD stages; GVHD 0 n = 2, GVHD I n = 1, GVHD II n = 4, GVHD IV n = 2 Maximum skin GVHD stages;GVHD 0 n = 2, GVHD I n = 4, GVHD II n = 1, GVHD III n = 1, GVHD IV n = 1 Maximun Gut GVHD stages; GVHD 0 n = 8, GVHD II n = 1	High-risk acute lymphoblastic leukemia, Wiskott-Aldrich Syndrome, HR-ALL, CNS 3, HR-ALL Philadelphia-like status-post-CD19 directed chimeric antigen receptor T cell therapy, X-Linked Lymphoproliferative Disease, DOCK 8 Deficiency	Cyclophosphamide, Cytarabine, total body irradiation, Busulfan, Fludarabine, ATG,	- Cefepime - Vancomycin - Piperacillin/tazobactam - Amoxicillin/clavulanic acid - Oxacillin	Within the first year the stage 3 0r 4 gut GVHD was occurred.	MRD, matched related donor; MSD, matched sibling donor, PBSC, peripheral blood stem cells; BM, bone marrow; UCB, umbilical cord blood; TBI, total body irradiation	16S rRNA Gene Sequencing
2-E Biagi/ 2019/Italy	36 pediatric patients	Male n = 20female n = 16	Stage I (n = 526.3 %), Stage II (n = 947.4 %), StageIII(n = 315.8 %), stageIV(n = 210.5 %), Only one patient experienced primary graft failure.During the procedure, 19 of 36 patients developed aGVHD; six had gut involvement.	ALL Acute Lymphoblastic Leukemia (n = 1438.9 %), AML Acute Myeloid Leukemia (n = 925.0 %), thalassemia major (n = 513.9 %). SAA Severe Aplastic Anemia (n = 38.3 %), BDA Blackfan-Diamond Anemia (n = 38.3 %), CDA Congenital Dyserythropoietic Anemia (n = 12.8 %), ALPS Autoimmune Lymphoproliferative Syndrome (n = 12.8 %)	Myeloablative(n = 3588.9 %), Nonmyeloablative(n = 111.1 %), - Myeloablative busulfan (or treosulfan)-based conditioning regimen n = 32 - Total body irradiation-based conditioning regimen n = 3 - Non-myeloablative preparative regimen with cyclophosphamide/ fludarabi n = 1	- Levofloxacin 5/36 - Piperacillin/tazobactam or Ceftazidime 29/36 - Cephazolin/piperacillin tazobactam 4/36	Mean day of onset (days from transplantation (+21)	29/36 patients received unmanipulated bone marrow cells, while 6/36 patients received T-cell depleted peripheral blood stem cells and 1/36 cord blood cells.	16S rRNA Gene Sequencing
3-E Biagi/ 2015/Italy	10 pediatric patients	Male n = 8, female n = 2	aGVHD grade I: n = 1, skin localization, aGVHD grade II: n = 2, skin localization, aGVHD grade III: n = 1, skin localization, aGVHD grade IV: n = 1, skin and gastrointestinal tract localization.	AML n = 2, ALL n = 6, Blackfan–Diamond anemia (BDA) n = 2	BU, TT = thiotepa, EDX = cyclophosphamide, Fludara = fludarabine, L-PAM = melphalan	NA	Grade I:+25 day, Grade II: +13 day, Grade III: +23 day, Grade IV: +11 day	Bone marrow n = 10	16S rRNA Gene Sequencing
4- Masetti/ 2023/Italy	90 pediatric	Male n = 53 female n = 37	NA	Acute lymphoblastic leukemia n = 33 (37 %), Acute myeloid leukemia n = 19(21 %), Non-Hodgkin lymphoma n = 2(2 %), MDS or JMML n = 10(11 %), Nonmalignant disease n = 26 (29 %)	Intensity of conditioning (%): Ablative n = 83(92 %), Reduced intensity n = 7 (8 %)	Ceftazidime or cefepime - CEF, piperacillin-tazobactam – P-T, and meropenem – MER	NA	Bone marrow n = 68 (76 %), PBSC n = 21 (23 %), Cord blood n = 1(1 %)	16S rRNA Gene Sequencing
5- Luo/2023/ China	22 children	Female = 12, male = 10	Stage I n = 3, Stage II n = 1, Stage IV n = 2	β-thalassemias major (β-TM)	All patients received busulfan (BU) + cyclophosphamide (CY) + anti-thymocyte globulin (ATG) as conditioning regime. Myeloablative intensity n = 22	NA	One patient developed gut GVHD on day 36 after infusion and was cured within 100 days. Four skin GVHD patients got	Peripheral blood n = 22	16S rRNA Gene Sequencing

(*continued on next page*)

Table 1 (*continued*)

First author/ Year/Country	Sample Size	Sex	Stage of GVHD	Indication for HSCT	Conditioning Regimen	Dose and type of antibiotics Intervention	HSCT Duration (Time to onset of GVHD)	Stem cell source	Methods of microbiome analysis
							the disease between 14 days and 1 month.		
6- B. Margolis/ 2023/USA	74 children	Female n = 31(41.9 %), male n = 43 (58.1 %)	None aGvHD: 61.5 %, aGvHD grade 1–2: 21 %, aGvHD grade 3–4: 17.5 %	Acute myeloid leukemia n = 44(59.5 %), Acute lymphoblastic leukemia n = 22(29.7 %), Hodgkin lymphoma n = 2(2.7 %), Myelodysplastic syndrome n = 2(2.7 %), Other leukemia or lymphoma n = 4(5.4 %)	Intensity conditioning; reduced intensity n = 49 (66.2 %), Myeloablative n = 24(32.4 %), Nonmyeloablative n = 1 (1.4 %). Conditioning radiation; None n = 31 (41.9 %), Total lymphoid irradiation n = 26(35.1 %), Total body irradiation n = 17(23.0 %)	NA	NA	Apheresis n = 47 (63.5 %), Marrow n = 27(36.5 %)	16S rRNA Gene Sequencing
7- Ingham/ 2021/ Denmark	29 children	Female = 13, male = 16	Nine patients (31 %) had no or mild aGVHD (grade 0 or I), and 20 patients (69 %) developed moderate to severe aGvHD (grade II–IV)	Malignant hematologic diseases(n = 20), Severe aplastic anemia (n = 1), Other benign disorders including immunodeficiencies(n = 8)	TBI + CY / TBI + VP16(n = 6), Combinations of BU, CY, VP16 and MEL(n = 6), Fludarabine-thiothepa combinations (n = 12), Fludarabine-thiothepa combinations(n = 5)	All patients were treated prophylactically with trimethoprim and sulfamethoxazole prior to HSCT. meropenem n = 28, vancomycin n = 24, ciprofloxacin n = 20, phenoxymethylpenicillin n = 14	Moderate to severe aGVHD (grade II–IV) at median + 14 days following HSCT (range: day +9 to day +61)	Bone marrow(n = 23), Umbilical cord blood(n = 2), Peripheral blood(n = 4)	16S rRNA Gene Sequencing
8- Song/2020/ China	42 pediatric patients	26 males,16 female	Six patients had Gut aGVHD; n = 0 had grade 0 diarrhea severity, n = 1 had grade I diarrhea severity, n = 3 had grade II diarrhea severity, n = 2 had grade III diarrhea severity.	11.36 % Acute lymphoblastic leukemia (ALL), 2.27 % Juvenile myelomonocytic leukemia (JMML), 4.55 % Myelodysplastic syndrome (MDS), 2.27 % Lymphoma, 2.27 % Neuroblastoma, 2.27 % Neuroblastoma, 2.27 % Fanconi anemia, 2.27 % Thalassemia, 15.91 % Mucopolysaccharidosis (MPS), 2.27 % Chronic granulomatous disease (CGD), 4.55 % X-linked high IgM syndrome (X-HIgM), 2.27 % Wiscot-Aldrich syndrome (WAS)	BU, busulfan; VP16, etoposide; CTX, cyclophosphamide; ATG, anti-thymocyte globulin; FLU, fludarabine; and TBI, total body irradiation, Methotrexate, cyclosporin	NA	NA	Peripheral blood with; MSD, HLA type matching sibling donor; MMRD, HLA type miss-matching related donor; MUD, HLA type matching unrelated donor; MMUD, HLA type miss-matching unrelated donor, Autologous	16S rRNA Gene Sequencing
9-M. Vossen/ 1990/ Netherlands	65 pediatric	Male = 23 Female = 21	aGVHD: acute 40 patients with complete gastrointestinal decontamination (GID); number of patients with > grade 2 actue GVHD; n = 0 in 11 patients with success of GID, n = 7 in 29 patients with not success of GID. 18 patients with selective GID; number	Severe aplastic anemia, Fanconi's anemia, MDS, leukemia, Acute non-lymphocytic, Acute lymphocytic leukemia, Chronic myelocystic leukemia	Cyclophosphamide, Procarbazine, total body irradiation, Methotrexate, Cyclosporin-A	For complete gastrointestinal decontamination: neomycin 1000 mg/day(For patients <2 years of age), 2000 mg/day (For patients >2 years of age), polymyxine B 1000 mg/day (For patients <2 years of age), 2000 mg/day (For patients >2 years of age), cephaloridine 1000 mg/day(For patients <2 years of age), 2000 mg/day (For patients >2 years of age), amphotericin B 1000 mg/day	NA	All patients received full bone marrow grafts from HLA genotypically identical siblings	16S rRNA Gene Sequencing

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Table 1 (*continued*)

First author/ Year/Country	Sample Size	Sex	Stage of GVHD	Indication for HSCT	Conditioning Regimen	Dose and type of antibiotics Intervention	HSCT Duration (Time to onset of GVHD)	Stem cell source	Methods of microbiome analysis
			of patients with > grade 2 actue GVHD; n = 6 in 11 patients with success of GID,n = 3 in 4 patients with not success of GID. cDVHD: chronic 35 patients with complete GID; 10 patients with success of GID(n = 2 have limited degree of cGVHD, n = 0 have Extensive degree of cGVHD), 25 patients with Not success of GID (n = 2 have limited degree of cGVHD, n = 3 have Extensive degree of cGVHD) 13 patients with selective GID; 11 patients with success of GID(n = 2 have limited degree of cGVHD, n = 3 have Extensive degree of cGVHD), 2 patients with Not success of GID (n = 1 have limited degree of cGVHD, n = 1 have Extensive degree of cGVHD)			(For patients <2 years of age), 2000 mg/day (For patients >2 years of age). For selective gastrointestinal decontamination: nalidixic acid 90 mg/kg, co-trimoxazole 12/ 60 mg/kg, polymyxin B 20 mg/ kg, neomaycin 15 mg/kg, amphotericin B 1000 mg/day (For patients <2 years of age), 2000 mg/day (For patients >2 years of age). 5-fluorocystosine			
10- M. Vossen1/ 2014/ Netherlands	112 pediatric	Male = 72, female = 40	The number of graft recipients that developed acute GVHD after BMT was very low, i.e. 9/112 (8 %). Eight children developed grade I GVHD according to the criteria of Przepiorka et al. [7], i.e. only involvement of the skin. One patient got grade II (skin stage 2, gut stage 1, and liver stage 2). None of the patients died as a result of acute GVHD.	High-risk acute leukemia = ALL:47,AML = 38 Chronic myelocytic leukemia = 3 juvenile myelo-monocytic leukemia = 7 Myelo-dysplastic syndrome = 12 non- Hodgkin lymphoma = 5	All children were pretreated with a myelo-ablative conditioning regimen This consisted of a combination of chemotherapy (cyclophosphamide, cytosin-arabinoside or VP-16 in some cases) and TBI. busulfan, busulfex. Prophylaxis consisted of MTX-short course) plus CsA.	Co-trimoxazole amphotericin B (2000 mg dd), gentamycin (800 mg dd) plus cefaloridin (2000 mg dd) cefuroxime or ceftriaxone IV in therapeutical dosage or by vancomycin per os (1000 mg dd) piperacillin/tazobactam was given per os (900 mg dd	NA	Bone marrow	NA

Table 2
Summary of results and limitations studies.

First author/ Year/Country	Results	Limitations
1-N. Gray/ 2023/USA	- Those experiencing stage 3–4 gut GVHD had significantly increased stool microbiota variation over time compared to control group ($p = 0.007$) - insignificant pre- HCT microbiota diversity with gut microbiota stability ($p = 0.05$)	- Data collection interruption due to Covid19 pandemic. - small sample size - different treatment courses between malignant and non-malignant pediatric patients; such as chemotherapy and antibiotic exposure - including pediatric patients under age 3 years old, therefore the gut microbiota did not stabilize.
2-E Biagi/ 2019/Italy	- Reduced diversity, lower Blautia content, increase in Fusobacterium abundance before HSCT caused developing gut aGVHD.	- Enrolment different hospital where patients treated in could impact the results.
3-E Biagi/ 2015/Italy	Indeed, in pre-HSCT samples, non-aGVHD patients showed higher abundances of propionate-producing Bacteroidetes. the aGVHD onset is associated, with a drop in the abundance of the known health-promoting Faecalibacterium and high percentages of Enterococcus.	- Difficult in fecal sample collecting - Small sample size - Reduced patients compliance during the illnesses.
4- Masetti/ 2023/Italy	The higher-diversity group had higher probability of overall surviica and lower incidence of grade 2 to 4 and grade 3 to 4 a GVHD. This group characterized by higher relative abundances of potentially health-related microbial families, such as Ruminococcaceae and Oscillospiraceae. In contrast, the lower-diversity group showed an overabundance of Enterococcaceae and Enterobacteriaceae. Network analysis detected short-chain fatty acid producers, such as Blautia, Faecalibacterium, Roseburia, and Bacteroides, as keystones in the higher-diversity group. Enterococcus, Escherichia-Shigella, and Enterobacter were instead the keystones detected in the lower-diversity group.	- Different centers with different clinical practicing - Small sample size - wide age range - Need linger and frequent sampling
5- Luo/2023/ China	The dominant bacteria were Bacteroidetes and Firmicutes at the phylum level and Lachnospiraceae at the family level before and after HSCT. In the differential analysis, Ruminococcaceae constantly decreased after HSCT. Besides, *Rothia mucilaginosa* was the most abundant 2 months after HSCT compared to before it. Additionally, GVHD patients presented decreased levels of Bacteroidetes compared to those without GVHD. Moreover, Blautia levels significantly decreased in critically ill GVHD patients	- Small sample size - Not considering the effect of die on gut microbiota
6- B. Margolis/ 2023/USA	The study found that ratios of Enterobacter species to oral microbes (Prevotella and Staphylococcus spp) were predictive of aGVHD (any grade) at baseline, while at neutrophil recovery a ratio of lactate producers (includes *Enterococcus dispar* and *Streptococcus sanguinis*) to strict anaerobes (includes Blautia massiliensis and *Bacteroides vulgatus*) was predictive of aGVHD (any grade), Arginine metabolism pathway presence at baseline was predictive of aGVHD (any grade)	- Large range age - Different conditioning regimen - different antibiotic exposure - different underlying disease - difficult stool sample containing - limited bacteria analysis
7- Ingham/ 2021/ Denmark	The study demonstrates that high abundances of intestinal *P. distasonis* (Tannerellaceae) as well as oral Actinomyces sp. (Actinomycetaceae) and other taxa from different body sites pre-HSCT predict the development of moderate to severe aGVHD post-transplant. When relating microbial abundances with immune cell counts, the study found rapid B and NK cell reconstitution to be associated with high abundances of Lachnospiraceae and Ruminococcaceae, which also depended on antibiotics treatment.	- -Antibiotics effects on microbiota composition
8- Song/2020/ China	The gut microbiota diversity was lowest when gut aGVHD occurred, which was consistent with the clinically higher mortality rate and greater treatment difficulty. Pasteurellaceae played an important role in gut aGVHD and diarrhea severity. Bifidobacteriaceae led to infectious diarrhea after HSCT. Specific bacteria were biomarkers for survival: Moraxellaceae, Enterococcaceae and Alphaproteobacteria from the intestinal flora after HSCT and Proteobacteria, Gammaproteobacteria and Odoribacteraceae before HSCT	- Small sample size - effect of underlying disease
9-M. Vossen/ 1990/ Netherlands	It is concluded that complete gastrointestinal decontamination in a strict protective environment is a feasible and very effective method for preventing severe infections and acute graft-versus-host disease after allogeneic bone marrow transplantation in children and adolescents; it resulted in a low transplantation-related mortality of 26 % and a good quality of survival in 69 % of the graft recipients.	NA
10- M. Vossen1/ 2014/ Netherlands	The results demonstrate that successful total GID of the graft recipient prevents moderate to severe acute GVHD. They suppose that substantial translocation of gastro-intestinal microorganisms or parts of these, functioning as microbial-associated molecular patterns	NA

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Table 2 (*continued*)

First author/ Year/Country	Results	Limitations
	(MAMP's), triggering macrophages/dendritic cells via pattern recognizing receptors (PRR's) is prohibited. As a consequence the initiation and progression of an inflammatory process leading to acute GVHD is inhibited.	

Table 3
Risk of bias assessment according to the Newcastle-Ottawa Quality Assessment Scale tool.

Study, Year (reference)	Selection	Comparability	Outcome/Exposure
1-N. Gray	***	*	***
2-E Biagi	****	**	**
3-E Biagi	**	**	**
4- Masetti	***	–	***
5- Luo	**	**	***
6- B. Margolis	****	*	**
7- Ingham	**	**	***
8- Song	**	**	***
9-M. Vossen	****	–	**
10- M. Vossen	***	*	***

mucilaginosa was the predominant bacterium two months post-HSCT compared to before, whereas *Ruminococcaceae* exhibited a consistent decline after HSCT. Margolis et al. [15]. demonstrated that the ratio of *Enterobacter* species to oral microbes (*Prevotella* and *Staphylococcus* species) was predictive of aGVHD at baseline, whereas at neutrophil recovery, the ratio of lactate producers to strict anaerobes was predictive of aGVHD. The presence of the arginine metabolic pathway at baseline was also predictive of aGVHD. The research conducted by Ingham et al. indicates that elevated levels of intestinal *P. distasonis (Tannerellaceae)*, together with oral *Actinomyces* sp. *(Actinomycetaceae)* and several taxa from other anatomical regions prior to HSCT, are predictive of the onset of moderate to severe aGvHD after transplantation [17]. The research indicates that bacterial diversity in the stomach, nose, and mouth decreased during the first month and then reestablished itself within 1 to 3 months post-allo-HSCT. In the 2023 research by Masetti et al. [7], the findings indicated that a higher-diversity group exhibited an increased chance of overall survival and a reduced incidence of grade 2 to 4 and grade 3 to 4 acute GVHD [7]. This group is distinguished by elevated relative abundances of possibly health-associated microbial families, including *Ruminococcaceae* and *Oscillospiraceae*. Conversely, the lower-diversity group had a surplus of *Enterococcaceae* and *Enterobacteriaceae*.

4. Discussion

The findings obtained from the study included in the present systematic study mostly indicated the use of combined treatments including BU, CY, and TBI and in some studies the use of ATG and MEL as conditioning regimens. In addition, out of 10 reviewed studies, 9 reported a significant decrease in gut microbiota diversity following GVHD. However, in all studies, an increased variation was reported. So that most of the studies showed a decrease in the levels of beneficial bacteria and producers of short-chain fatty acid products in the intestine such as *Ruminococcaceae* and *Enterococcus*, which is also observed in the intestinal microbiota population of healthy people [11,15,20]. *Enterococcus* is indeed one of the producers of short-chain fatty acids (SCFAs), particularly butyrate, which is a key SCFA known for its beneficial effects on gut health. While *Enterococcus* species are primarily known for their role in the gastrointestinal tract, some strains are capable of fermenting dietary fibers and producing SCFAs as by-products of this fermentation [22].

The onset of aGVHD has been correlated with the gut microbiota (GM) composition, however, studies investigating this association involve mainly the adult population and only a limited number of studies have been performed involving pediatric patients [11,7,14–21]. Patients undergoing allogeneic hematopoietic cell transplantation often have gastrointestinal complications influenced by aGVHD. The etiology of this gastrointestinal disturbance is linked to a dysbiotic microbiome, and since the microbial composition changes throughout the gut length, the microbiome may correspond with the pattern of organ involvement after allo-HSCT [23].

A meta-analysis involving adult patients demonstrated that a higher intestinal microbiota diversity was significantly associated with an overall survival benefit, decreased risk of transplant-related mortality and lower incidence of grade II-IV aGVHD. This study also showed that higher levels of *Clostridiales* were associated with superior overall survival, while the higher abundance of *Enterococcus*, *γ-proteobacteria,* and *Candida* was a poor overall survival [24]. Increased gut microbiota diversity correlated with improved outcomes throughout both pre- and post-transplant phases. A further meta-analysis indicated a decrease in grade II to IV acute and intestinal GVHD among individuals who did not undergo gut decontamination [25]. Moreover, Taur et al. demonstrated that adult patients with decreased bacterial diversity in stool seven days after neutrophil engraftment had a worse three-year survival rate compared to those with increased intestinal diversity at the same time point (36 % vs. 67 %) [26]. Peled et al. have shown that diminished intestinal diversity correlates with increased mortality due to transplantation and gut GVHD in adults [27]. These findings indicate a crucial role of gut microbiota in influencing mortality outcomes after HSCT. Biagi et al. revealed that children with acute GVHD have significant bacterial gut dysbiosis during the first 3–4 months after allogeneic HCT, unlike those without GVHD [11]. Furthermore, children with GVHD exhibited a diminished prevalence of *Bacteroidetes*, a phylum known for its synthesis of propionic acid, which has been demonstrated to improve gut integrity and promote bacterial colonization in naïve hosts, thus contributing to the compromised or delayed mucosal repair observed in GVHD patients. Elgarten et al. [10] discovered that children receiving allogeneic hematopoietic cell transplantation had a pre-transplant stool composition characterized by a reduction in *Bacteroides* and *Ruminococcaceae* compared to healthy controls, possibly attributable to pre-transplant chemotherapy and/or antibiotic exposure. Moreover, Ingham et al. [17] corroborated the findings of Elgarten et al. by demonstrating that the microbial composition preceding HSCT could predict the severity of acute GVHD post-transplantation using a machine learning approach; however, longitudinal biospecimens are necessary to validate these results. Pre-HSCT stool composition may act as a significant predictor for identifying patients at increased risk of developing severe gut GVHD.

The gut microbiome is linked to GVHD, according to common results from these significant investigations. However, findings may vary among transplant centers due to variations in antibiotic-use practices, individual antibiotic histories, hosts, treatment protocols, and other unmeasured factors. In order for the broader community of patients receiving allogeneic HSCT to benefit from these scientific results, we must establish common patterns and processes that relate the microbiome to the risk of GVHD. For instance, almost all research links a lower variety of stool bacteria to a higher risk of GVHD or death. Numerous studies have linked enterococci to an increased incidence of GVHD; however, it is unclear how lactobacilli contribute to GVHD [28,29]. In contrast to people lacking lactobacilli, it is likely that lactobacilli may rise in concentration prior to the beginning of GVHD while still providing some protection. This scenario is supported by Jenq and colleagues' results from a mouse model [28] which show that whereas microorganisms may be indicators of GVHD risk, modulators of GVHD risk, or both, the directionality of these connections might be complicated.

Tong et al. (2024) [30] examine the distribution of intestinal flora after HSCT in children, finding a significant alteration in gut microbiota, particularly a reduction in beneficial anaerobic bacteria such as Clostridium and Bacteroides, which may influence patient recovery and susceptibility to infections. Similarly, Faraci et al. (2024) [31] investigate the association between oral and fecal microbiome dysbiosis and treatment complications, reporting that microbiome imbalances are linked to increased risks of gastrointestinal toxicity, infections, and graft-versus-host disease (GVHD). Both studies emphasize the importance of monitoring and modulating the microbiome to predict and mitigate complications in pediatric HSCT recipients.

Maintaining microbiota diversity is thought to be beneficial in preventing severe intestinal GVHD after HSCT. Reduced gut microbiota diversity is associated with more severe GVHD, likely due to impaired immune regulation. Restoring microbiota diversity through interventions like fecal microbiota transplantation (FMT) could help reduce GVHD severity and improve patient outcomes [24,30]. While reduced microbiota diversity is linked to poor transplant outcomes, it remains unclear whether this is a causative factor or just an epidemiological marker. Some studies suggest that low diversity directly affects immune function and inflammation, while others argue it may simply reflect the patient's clinical state [17,23,30]. Further research is needed to clarify whether microbiota diversity influences disease progression or is a consequence of underlying conditions.

Furthermore, antibiotic exposure during the transplant period may affect a patient's ability to maintain or restore microbial diversity before HSCT, according to the majority of studies included in our review. Given that reduced microbiota diversity after HSCT correlates with GVHD and transplantation-related mortality, prudent antibiotic administration throughout the pre- and peri-engraftment phases is essential [14]. In the HSCT population, gastrointestinal GVHD may be harmful. It has been shown that the alteration of the microbiome brought on by the use of antibiotic therapy—particularly broad-spectrum antibiotics—increases both acute GI GVHD and GVHD-associated mortality [32]. Although it is advised to take antibiotics sparingly, overuse of them during neutropenic fevers may result in sepsis or even death. Thus, it is necessary to balance the advantages and disadvantages of early antibiotic treatment. Research on the de-escalation of broad-spectrum antibiotics in cancer patients with fever of unclear origin and febrile neutropenia was examined by Pillinger and colleagues (2020) [33]. They found evidence in favor of de-escalation without a difference in recurring fever. Additionally, since the gut microbiome is so extensive, additional research is need to look at how antibiotics may affect different species that might damage the gut flora and cause gastrointestinal GVHD on a wider scale. The impact of diet on gut diversity throughout the transplantation process and the effects of fecal transplants on the gut microbiota in gastrointestinal GVHD patients are two other potential research topics.

Conditioning regimens in autologous HSCT have a significant impact on the gut microbiota. These regimens, which typically involve chemotherapy and/or radiation, can lead to a profound shift in the microbial composition, reducing microbial diversity and promoting dysbiosis. The destruction of gut epithelial cells and the disruption of mucosal barriers during conditioning may allow pathogenic bacteria to proliferate, potentially leading to infections or complications such as GVHD. Studies have shown that the intensity of conditioning regimens correlates with the extent of microbiota disruption, with more aggressive treatments causing greater microbial shifts. Restoring microbiota diversity after conditioning may help mitigate complications and improve outcomes, suggesting the potential for microbiome-targeted therapies post-transplant [6,8,27,30].

The role of the gut microbiota in the development of GVHD has been explained by a number of different mechanisms. By allowing microbes or microbe-associated molecular patterns (MAMPs) to penetrate an epithelial barrier that has been damaged by conditioning chemotherapy and radiation, the microbiota may promote host cell inflammation [34,35]. Following their interaction with pattern recognition receptors on human cells, these bacteria and MAMPs may attract inflammatory cells involved in GVHD. Additionally, via the synthesis of compounds such SCFA butyrate (e.g., by Lachnospiraceae), the gut microbiota might affect host cell physiology [36]. In addition to serving as intestinal epithelial cells' main energy source, butyrate also affects inflammatory signaling pathways [37] and, at high concentrations, promotes intestinal wound healing [38]. Low butyrate levels in intestinal tissue were linked to GVHD [39], and SCFAs produced by the gut microbiota in a mouse model reduced GVHD [40]. Additionally, regardless of Treg number, gavage with butyrate or a Clostridium combination decreased GVHD [39]. Additionally, butyrate inhibits antigen-stimulated T cells that induce GVHD by acting as a histone deacetylase inhibitor [41,42]. These findings point to a molecular connection between the risk of GVHD and gut metabolites and microorganisms.

This study had limitations, including that each study analyzed the microbiome using different methods, such as surveillance cultures and 16S rRNA sequencing, which prevented us from conducting meta-analyses. In addition, microbiome diversity was assessed using different metrics, including the Shannon index, the inverse Simpson index, and OTU counts. They also used different methods to assess microbiome diversity and used different thresholds to distinguish between low and high diversity, which was another limitation of this study. It is also well known that the composition of a normal/healthy microbiome varies significantly with age in pediatric populations, reflecting differences in dietary stages, such as between breastfed and non-breastfed infants. This variability makes microbiome research in pediatric populations challenging. However, most studies have accounted for and considered these factors as confounders.

4.1. Conclusion

As a result, our findings indicated a possible decrease in diversity as well as a change in intestinal microbiota in children with GVHD under HSCT in most of the studies. Our results provide a comprehensive profile of pediatric patients post-allogeneic HSCT and may inform bigger prospective research aimed at elucidating the variables that influence a child's capacity to maintain a stable and diversified gut microbial population after HSCT. Also, these data can be used to adjust the preparation conditions and antibiotic therapy with effective interventions to improve or prevent GVHD.

Authors' contributions

P.R and Mh.S contributed in conception, design, and statistical analysis. A.Z., Z.H, P.R, M.B, A.H, and MH·S contributed in data collection and manuscript drafting. P.R and A.H supervised the study. All authors approved the final version of the manuscript.

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Declaration of competing interest

We, the authors, declare that we had no competing interests.

Availability of data and materials

Data available on request due to privacy/ethical restrictions.

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